WW-P-377d
November 5, 1962

SUPERSEDING
Int. Fed. Spec. WW-P-00377c (Navy-Ships)
June 23, 1961 and
Fed. Spec. WW-P-377b
April 19, 1954

FEDERAL SPECIFICATION

PIPE, COPPER, SEAMLESS, STANDARD SIZES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

This specification covers seamless copper pipe, in all standard pipe sizes, regular weight only.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 146—Tolerances for Copper and Copper Base Alloy Mill Products.

Fed. Test Method Std. No. 151—Metals; Test Methods.

Fed. Std. No. 185—Identification Marking of Copper and Copper Base Alloy Mill Products.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-C-3993—Copper and Copper-Base Mill Products; Packaging of.

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless other-

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wise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 **Chemical requirements** The material shall conform to the chemical requirements shown in table I.

TABLE I.—*Chemical requirements*

Phosphorous (maximum)	Copper[1] (minimum)
Percent	*Percent*
0.040	99.90

[1] Silver counting as copper.

3.2 **Temper.** Unless otherwise specified (see 6.2), the pipe shall be furnished in the hard-drawn temper. When pipe is required for bending, pipe shall be furnished in the temper as specified (see 6.2).

3.3 **Mechanical properties.**

3.3.1 *Bending.* When pipe is required for bending (see 6.2), the pipe 2 inches and under in outside diameter shall withstand the bend test specified in 4.4.2.

3.3.2 *Expansion.* The pipe shall withstand the expansion test specified in 4.4.3 or, as an alternative, the flattening test specified in 4.4.3.1.

3.4 **Microstructure.** The pipe shall be free of cuprous oxide when examined as specified in 4.4.5.

3.5 **Hydrostatic-test pressure.** Hydrostatic-test pressure shall be as specified in table II, when tested as specified in 4.4.1 (see 6.5).

TABLE II.—*Hydrostatic-test pressure*

Standard pipe size	Hydrostatic-test pressure per square inch (minimum)
Inches	*Pounds*
1/8	1,000
1/4	1,000
3/8	1,000
1/2	1,000
3/4	1,000
1	1,000
1 1/4	1,000
1 1/2	1,000
2	900
2 1/2	900
3	900
3 1/2	900
4	800
5	600
6	500
8	500
10	400
12	350

3.6 **Tolerances.** Tolerances shall not exceed that specified in the following sections of Fed. Std. No. 146:

(*a*) Dimensions and weight, standard pipe size (copper).

(*b*) Weight and wall thickness tolerances (copper).

(*c*) Length tolerances (copper).

(*d*) Squareness of cut (copper).

(*e*) Roundness tolerances (copper).

3.6.1 When specified (see 6.2), tolerance shall be all plus or all minus.

3.7 **Size of pipe.** The size of pipe required shall be as specified (see table II and 6.2).

3.8 **Marking.** When specified (see 6.2), identification marking shall be in accordance with Fed. Std. No. 185.

3.9 **Workmanship.** The pipe shall be clean, smooth, round, straight, of the proper dimensions, and with ends cut square. It shall be free from injurious grooving, indentations, cracks, flaws, slivers, spilly me-

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tal, or other harmful defects which would interfere with the use of the material in the application for which it is intended.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the government as specified in the contract or order. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection

4.2.1 *Inspection lot.* For the purpose of inspection, a lot shall consist of all pipe of the size in the following quantities.

Nominal pipe size	Weight of lot
Inches	*Pounds*
Up to 1 1/2, inclusive	5,000
Over 1 1/2 to 4, inclusive	10,000
Over 4	20,000

4.2.2 Samples selected for purposes of the tests prescribed in this specification shall be selected in a manner as to represent correctly the material furnished. Samples for tests shall be selected in accordance with table III.

TABLE III.—*Sampling for destructive tests*

Number of pieces in the lot	Number of sample pieces to be taken
1 to 50	1
51 to 200	2
201 to 1500	3
Over 1500	0.2 percent of total number of pieces in the lot but not to exceed 10 sample pieces

4.2.3 *Sampling for examination.* A random sample of pipe shall be taken in accordance with table IV from each inspection lot of material offered for examination of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with MIL-STD-105.

4.2.4 *Sampling for bending and expansion* tests. Samples selected as specified in 4.2.2 shall be subjected to the bending and expansion tests specified in 4.4.2 and 4.4.3.

4.2.5 *Sampling for chemical analysis.* Two ounces of clean chips shall be taken from each sample selected as specified in 4.2.2 for chemical analysis.

4.3 Examination.

TABLE IV.—*Sampling for visual and dimensional examination AQL (approx.) = 1.5 percent defect*

Number of pipes in an inspection lot	Number of pipes in sample	Acceptance number (defective)	Rejection number (defectives)
40 or under	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1300	110	4	5
1301 and over	150	5	6

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4.3.1 *Visual and dimensional examination.* Each sample pipe taken in accordance with table IV shall be visually and dimensionally examined to verify compliance with this specification. Any pipe in the sample containing one or more visual and dimensional defects shall not be offered for delivery, and if the number of defective pipes in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.3.2 *Examination of preparation for delivery.* The packaging, packing, and marking shall be examined to determine compliance with the requirements of section 5 of this specification.

4.4 **Tests.**

4.4.1 *Hydrostatic-pressure test.* Each pipe, in the temper in which it is to be delivered, shall be tight under pressure and shall show no bulges, cracks, flaws, porous places, or other harmful defects when tested to the minimum internal hydrostatic pressure specified in table II.

4.4.1.1 In lieu of the hydrostatic-pressure test specified in 4.4.1, the pipe may be tested under water with air at 60 to 100 pounds per square inch internal pressure.

4.4.2 *Bending test.* When required (see 3.3.1), annealed full sections of pipe, 2 inches and under in outside diameter, shall be bent cold 180 degrees around a pin, the diameter of which is one and one-half times the inside diameter of the pipe, without cracking on the outside of the bent portion.

4.4.3 *Expansion test.* Specimens selected for the test shall, after annealing, withstand an expansion of 25 percent of the inside diameter when expanded by a tapered pin having a 60-degree angle (see 3.3.2) and the expanded area shall show no cracks or ruptures visible to the unaided eye.

4.4.3.1 *Flattening Test.* In lieu of the test specified in 4.4.3 for pipe over 4 inches in diameter in the annealed condition, a section 4 inches in length shall be selected as specified in 4.2.2. The specimen shall be flattened so that a gage set at three times the wall thickness will pass over the pipe freely throughout the flattened part. The pipe so tested shall develop no cracks or ruptures visible to the unaided eye. In making this test the elements shall be slowly flattened by one stroke of the press.

4.4.4 *Chemical analysis.* Chemical analysis shall be made by the wet chemical method in accordance with method 111 of Fed. Test Method Std. No. 151 or by the spectrochemical method in accordance with method 112 of Fed. Test Method Std. No. 151. In case of dispute the chemical analysis by wet chemical methods shall be the basis for acceptance.

4.4.5 *Miscroscopic examination.* Microscopic examination of pipe specimens selected as specified in 4.2.2 shall be made at a magnification of 75X to determine conformance with 3.4.

4.5 **Rejection.** If any specimen fails to conform to this specification, this shall be cause for the rejection of the entire lot, except as specified in 4.3.1 and 4.4.1, subject to the retest provisions in accordance with Fed. Test Method Std. No. 151 and the disposition of nonconforming product in accordance with MIL-STD-105.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* **Packaging of piping shall be in accordance with Level A of MIL-C-3993.**

5.1.2 *Level C.* **Packaging of piping shall be in accordance with the supplier's commercial practice.**

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5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Piping packaged as specified (see 5.1) shall be packed in accordance with level A of MIL-C-3993.

5.2.2 *Level B.* Piping, packaged as specified (see 5.1), shall be packed in accordance with level B of MIL-C-3993.

5.2.3 *Level C.* Piping, packaged as specified (see 5.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers or method of packing shall be in accordance with the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Military agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-C-3993.

5.3.2 *Civil agencies.* Marking for shipment shall be in accordance with the standard marking of Fed. Std. No. 123.

6. NOTES

6.1 **Intended use.** Seamless copper pipe is used for plumbing, boiler feed lines, and similar purposes.

6.2 **Ordering data.** Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Temper, if other than hard drawn (see 3.2).
(*c*) Temper to be furnished when pipe is required for bending (see 3.2).
(*d*) If bending is required (see 3.3.1).
(*e*) If tolerance is all plus or all minus see 3.6.1).
(*f*) Pipe size required (see 3.7). Sizes other than those listed in table II may be purchased.
(*g*) Total number of feet of each size required.
(*h*) Marking, if required (see 3.8).
(*i*) Levels of packaging and level of packing required (see 5.1 and 5.2).
(*j*) Special marking for shipment required (see 5.3.1).

6.3 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.4 The chemical and mechanical property requirements of this specification are similar to ASTM B42-58.

6.5 **Hydrostatic pressure.** For sizes not listed in table II, internal hydrostatic test pressure for thin, hollow cylinders should be as follows:

$$P = \frac{2\ St}{D-.8t}$$

in which:

P=safe internal pressure per square inch.
D=outside diameter in inches.
S=allowable unit stress of the material, 6,000 pounds per square inch.
t=wall thickness in inches.

6.6 The requirement for item identification marking specified in 3.8 and packaging, packing, and marking for shipment specified in section 5 apply to direct shipment for Government activities and apply also where specified to contract or orders between the manufacturer and the government prime contractor.

6.7 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

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Rail:

Pipe, copper, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Pipe, copper, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification

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Copies of this specification may be purchased for 5 cents each.

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